Can Reinforcement Learning Solve Asymmetric Combinatorial-Continuous Zero-Sum Games?

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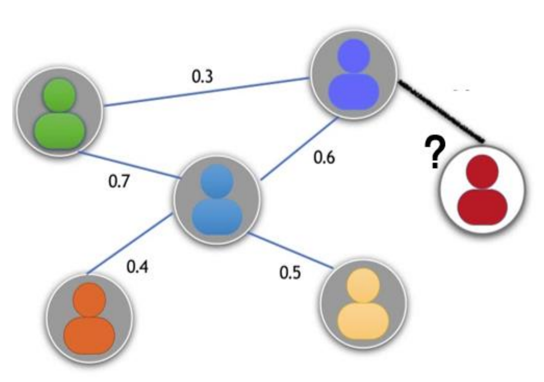
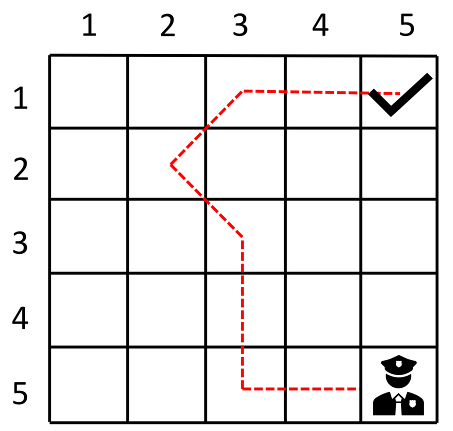
What's Asymmetric Combinatorial-Continuous zEro-Sum (ACCES) games?

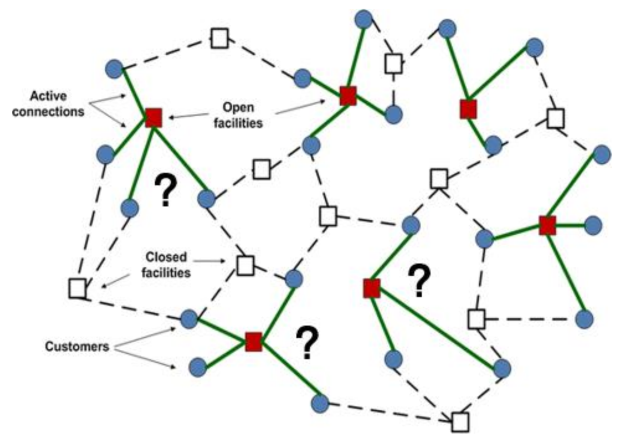
- Two-player zero-sum game with dynamics of simultaneous move and static form.
- Player 1 has a combinatorial strategy space and player 2 owns an infinite and compact strategy space with a continuous utility function.



IM with uncertain weights



Patrolling game



Facility location with unknown effect

Challenges in ACCES Games

- Does the Nash Equilibrium (**NE**) **exist**?
- Is there any **algorithm** that can **converge to NE**?
- Is there a **practical algorithm** that we can actually **implement in the real world**?

Contributions

- First paper to summarize and define ACCES games.
- Prove the **existence of mixed NE** in ACCES games.
- Propose the **CCDO** and **CCDO-RL** framework:
  - Novel convergence guarantee of two algorithms.
  - CCDO-RL: First practical algorithm to solve ACCES games.

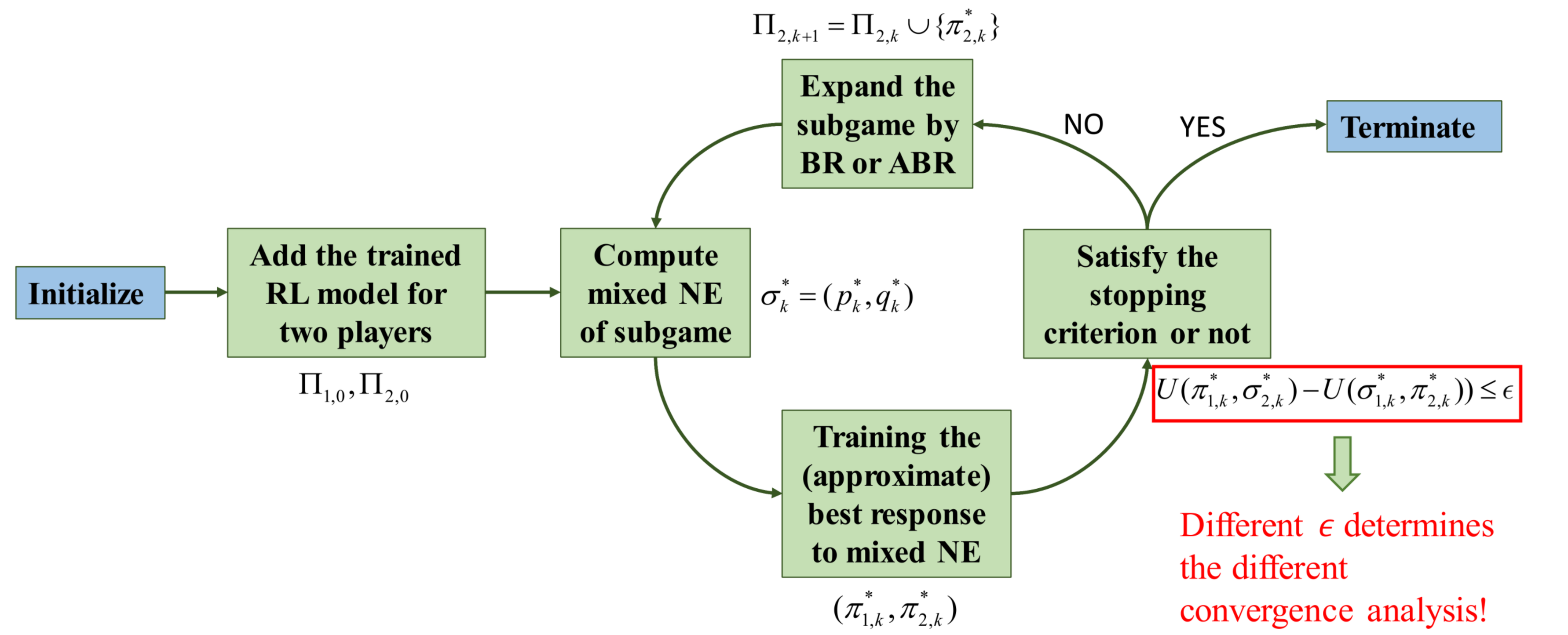
Algorithms

Algorithm 1 Combinatorial-Continuous Double Oracle Reinforcement Learning Algorithm

Input: Game $\mathcal{G}=(X,Y,u)$, X,Y are the pure strategy spaces for Player 1 and 2, respectively. Stopping criterion $\epsilon \geqslant 0$.

Output: σ_k^*, the mixed equilibrium of the subgame.

Initialize strategy set $\Pi_{1,0},\Pi_{2,0}$ of Player 1 and 2 respectively.
repeat
  Solve the mixed equilibrium $\sigma_k^*=(\sigma_{1,k}^*,\sigma_{2,k}^*)$ in the subgame $(\Pi_{1,k},\Pi_{2,k})$.
  Find the approximate best response (ABR) by RL algorithms $(\pi_{1,k}^*,\pi_{2,k}^*)$.
  $\Pi_{1,k+1}=\Pi_{1,k}\cup\{\pi_{1,k}^*\}$, $\Pi_{2,k+1}=\Pi_{2,k}\cup\{\pi_{2,k}^*\}$.
  if $U(\pi_{1,k}^*,\sigma_{2,k}^*)\leqslant U(\sigma_k^*)$ **then** ▷ U is the expected utility function
    $\Pi_{1,k+1}=\Pi_{1,k}$. ▷ Delete Player 1's ABR $\pi_{1,k}^*$ from $\Pi_{1,k+1}$
  else if $U(\sigma_{1,k}^*,\pi_{2,k}^*)\geqslant U(\sigma_k^*)$ **then**
    $\Pi_{2,k+1}=\Pi_{2,k}$. ▷ Delete Player 2's ABR $\pi_{2,k}^*$ from $\Pi_{2,k+1}$
  end if
until $U(\pi_{1,k}^*,\sigma_{2,k}^*)-U(\sigma_{1,k}^*,\pi_{2,k}^*))\leqslant\epsilon$



CCDO Convergence Analysis (Theorem 2):

1. When the stopping criterion $\boldsymbol{\epsilon}=\mathbf{0}$, CCDO possibly iterates in an infinite number of iterations. However, every weakly convergent subsequence in the **subgame equilibrium sequence** $\{p_k^*,q_k^*\}$ converges to **the equilibrium of the whole game**.
2. When the stopping criterion $\boldsymbol{\epsilon}>\mathbf{0}$, CCDO converges to an **ϵ-equilibrium** in a finite number of epochs.

CCDO-RL Convergence Analysis (Theorem 3):

1. When stopping criterion $\boldsymbol{\epsilon}=\mathbf{0}$, if the approximate response oracle for Player 2 has a **uniform lower bound** for every mixed strategy, CCDO-RL must converge to an **$(\epsilon+\epsilon_1+\epsilon_2)$-equilibrium** in a finite iterations.
2. When stopping criterion $\boldsymbol{\epsilon}=\mathbf{0}$ and CCDO-RL iterates **infinite** rounds, every weakly convergent subsequence converges to an **ϵ_1- equilibrium**.
3. When the stopping criterion $\boldsymbol{\epsilon}>\mathbf{0}$, CCDO-RL converges to an **$(\epsilon+\epsilon_1+\epsilon_2)$ -equilibrium** in a finite number of epochs.

Experimental Results

Exploitability metric:

(From left to right:

ACSP, ACVRP, PG)

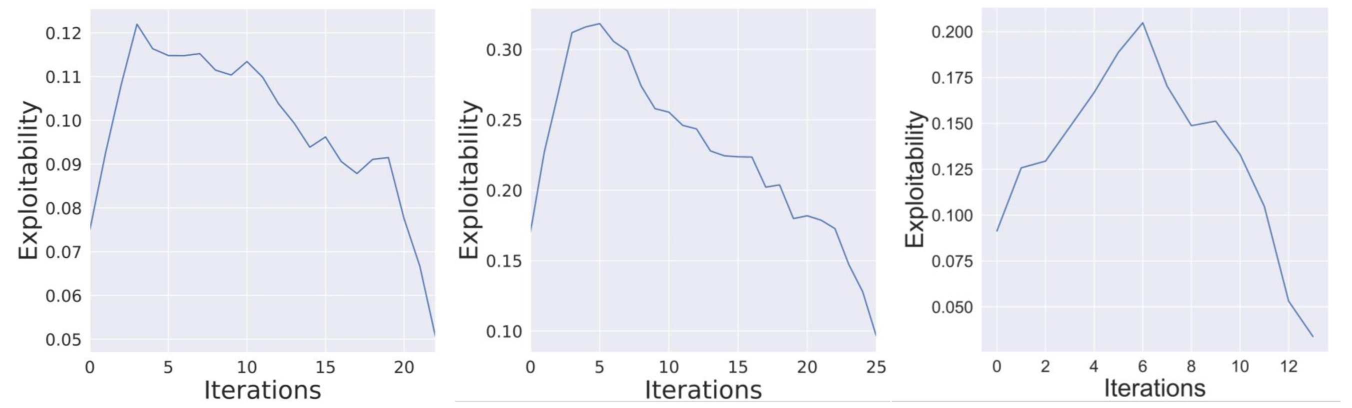


Table 1: Average reward against CCDO-RL's adversary (on seen graphs)

method	ACSP (Mean±Std)		ACVRP (Mean±Std)		PG (Mean±Std)	
	20 nodes	50 nodes	20 nodes	50 nodes	20 nodes	50 nodes
Heuristic	6.13±1.20	7.55±1.42	7.65±1.23	13.38±1.70	2.64±1.03	4.53±1.84
RL against Stoc	3.50±0.47	4.55±0.62	7.55±1.16	13.90±1.63	2.71±0.90	4.80±2.18
CCDO-RL	**3.25**±0.42	**4.31**±0.51	**7.42**±1.21	**13.28**±1.52	**2.75**±0.87	**5.01**±1.91

Table 2: Generalizability against CCDO-RL's adversary (on unseen graphs)

method	ACSP (Mean±Std)		ACVRP (Mean±Std)		PG (Mean±Std)	
	20 nodes	50 nodes	20 nodes	50 nodes	20 nodes	50 nodes
Heuristic	6.20±1.33	7.60±1.37	7.64±1.30	13.27±1.87	2.43±0.98	4.19±1.69
RL against Stoc	3.56±0.37	4.57±0.58	7.67±1.30	13.85±1.53	2.50±0.95	4.26±2.17
CCDO-RL	**3.31**±0.35	**4.39**±0.52	**7.55**±1.28	**13.15**±1.59	**2.56**±0.92	**4.70**±1.94

[1] For the average reward of ACSP and ACVRP, smaller is better while for that of PG larger is better.